

AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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PROFESSIONAL SUMMARY

State Engineer graduated from ESI-SBA, specializing in distributed architectures, generative surrogates, and metaheuristic optimization. Expert in cyber-physical systems (CPS) and edge-cloud inference for energy systems. PhD candidate focused on Distributed AI for collaborative energy performance.

WORK EXPERIENCE

AI Engineer Intern — Digital Twin Edge Computing Jan 2026 – Present
Sonatrach (Pipeline Transport Branch)

- Designed a dual Edge/Cloud architecture for industrial gas turbine monitoring.
- Achieved **0.04ms** edge inference latency using **C++ JIT Compiled xLSTM**.
- Developed a generative surrogate based on **Mamba-SSM** for "What-If" simulations.
- Integrated **Physics-Informed Neural Networks (PINN)** for thermodynamic consistency.

Network Engineering Intern Sep 2024
Algérie Télécom RMC Center

EDUCATION

National School of Computer Science (ESI-SBA, Algeria) 2021 – 2026
State Engineer's Degree & MSc in Artificial Intelligence

TECHNICAL SKILLS

- Distributed AI:** Federated Learning (architecture design), Distributed Inference, Flower.
- Optimization:** PSO, Genetic Algorithms (GA), ACO, Bayesian Optimization.
- Deep Learning:** PyTorch, Mamba-SSM, xLSTM, Transformers, GNN, PINN.
- Stack:** C++, Python (FastAPI), Docker, Kubernetes, MLflow, CUDA, SQL.

RESEARCH ENGINEERING PROJECTS

University Scheduling (NP-Complete Optimization) 2025
Automated timetabling system using hybrid metaheuristics (**PSO/ACO/GA**). *Key skill: Dynamic optimization under hard constraints.*

Aura — Adaptive Reinforcement Learning System 2025
Trading platform with distributed real-time data pipelines and DQN agents.

Geo-RAG — Geospatial Knowledge Retrieval 2025
Intelligent geospatial assistant using RAG and spatial reasoning for decision support.

Advanced NLP Optimization 2025
Multilingual content moderation optimized via metaheuristic hyperparameter search.

Industrial Safety Tracking (CV) 2025
Real-time PPE detection and object tracking in industrial environments (YOLOv8).

LANGUAGES

Arabic (Native), English (Advanced), French (Fluent).